

2021 Final round of Nepal Mathematics Olympiad

October 30, 2022

Problem 1

Find all functions $f : R \rightarrow R$ such that $f(x + y) \leq f(x) + f(y)$, $f(x) \leq x$

Source

Problem 2

If n is not divisible by 5, prove that we can divide angle $\frac{\pi}{n}$ into 5 equal parts using compass and ruler.

Source

Problem 3

Imagine 2021 boxes in a row and a hidden bird inside one of them. We want to hunt the bird with a shotgun such that we can shot at any of the boxes in each step. But as we shot the bird goes into the left or right neighbour box of where it was before (if exists) because of shot sound. Prove that we can hunt the bird with less than 5000 shots.

Source

Problem 4

A simple curve lies inside a circle such that it bisects the area of circle and it connects two arbitrary points of the circle. Prove that the length of the curve is more than or equal to the diameter of the circle.